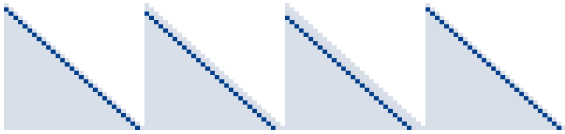
Attention Layer 1

(a) Time 600

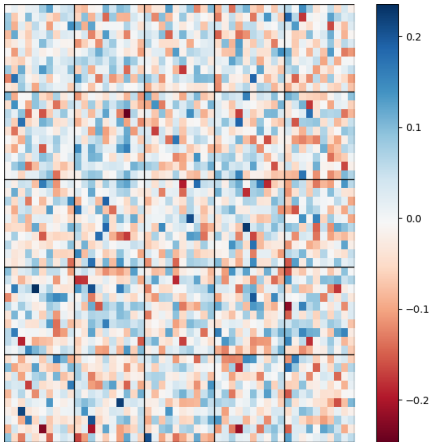
(b) Time 1100

(c) Time 1600

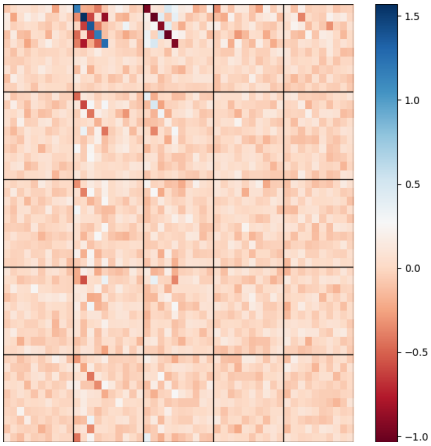


(d) Time 2300

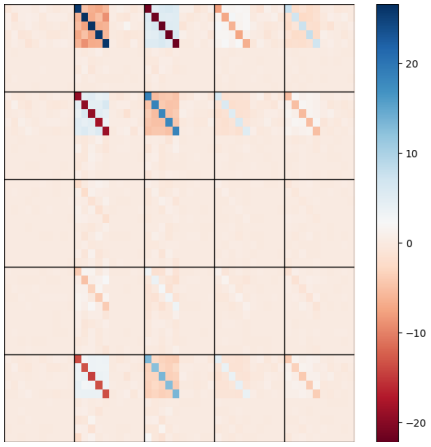
$Q^\top K$ Layer 2



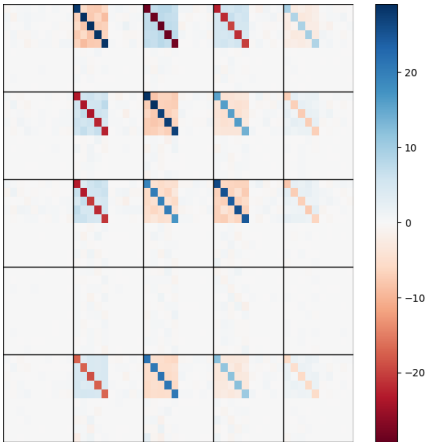
(e) Time 600



(f) Time 1100



(g) Time 1600



(h) Time 2300

Figure 1: **Q,K in second layer with random initialization.** Plots for $k = 5$ and $l = 120$ without assumption A1, A2. We use Gaussian initialization for Q and K with variance 0.1, making $Q^\top K$ random at initialization as is evident from time 600. We see the evolution of the second layer where the top principal diagonal of the block matrix aligns with $\tilde{I}$. Although other blocks also move, the main principal diagonal stays close to our simplification.

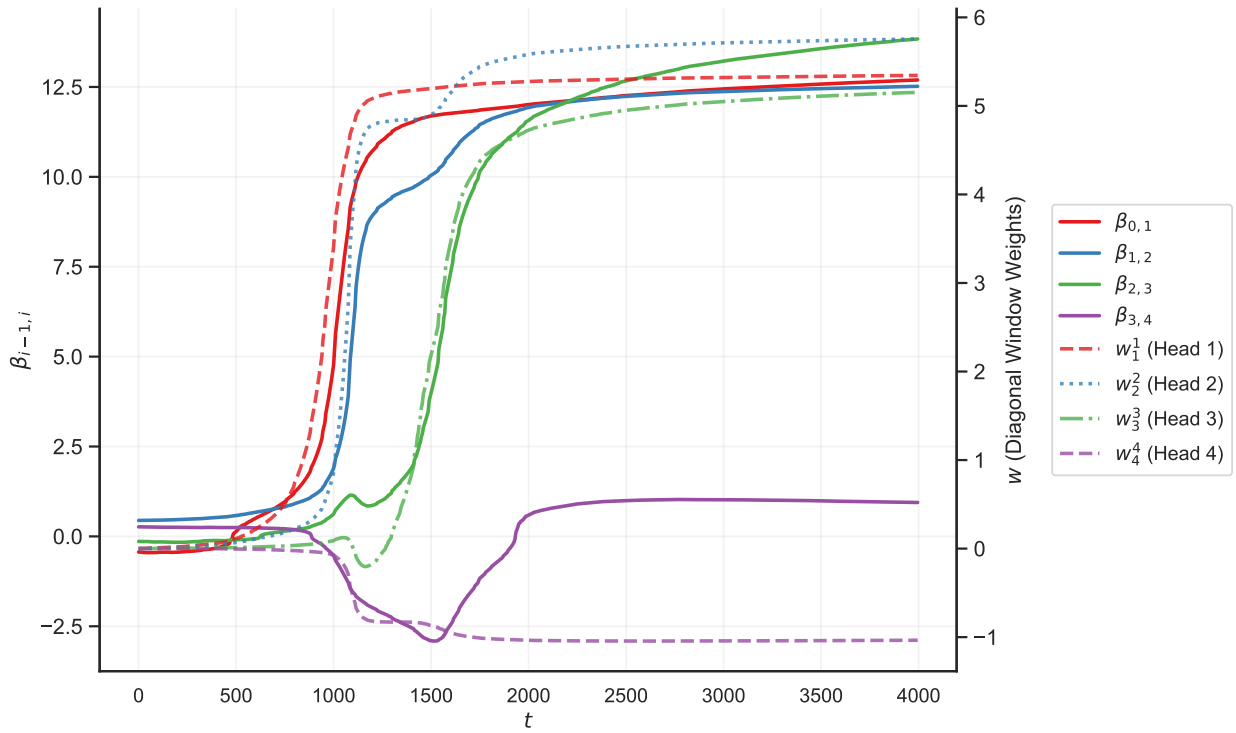


Figure 2: **Betas and Window Weights with full Q,K initialization.** Again for the same setting as Figure 1. We plot the attention weights and also the betas. We compute the betas by taking the inner product of the blocks of $Q^\top K$ with I. We see that the betas, attention weights are tied and evolve in a way that is consistent with our conservative law and theory. The triaing is not complete as it is taking a long time to leave the last plateau.

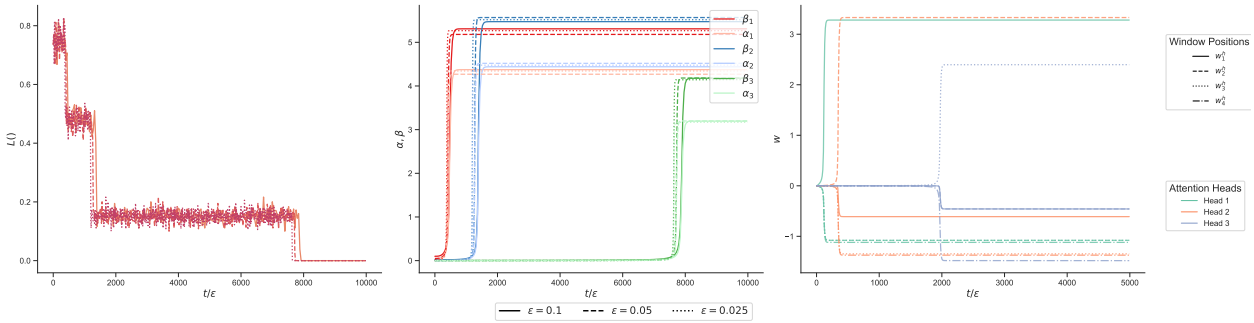


Figure 3: Incomplete permutations for $k = 4$ and length $= 60$ at various scales trained with cross entropy and with assumptions A(0)-A(4). It validates the timescales and conservative law when all the permutations are not present in the context.